2.2 Classwork: Average and Instantaneous Velocity Solutions

1. Swimming.

   (a) Average speed $= \dfrac{50\text{ m}}{20.91\text{ s}} = 2.3912\ldots \approx \mathbf{2.391\ m/s}$. Average velocity $\approx$ **2.391 m/s down the pool**.

   (b) Average speed $= \dfrac{100\text{ m}}{46.40\text{ s}} = 2.1551\ldots \approx \mathbf{2.155\ m/s}$. Average velocity $= \dfrac{0\text{ m}}{46.40\text{ s}} = \mathbf{0}$.

   In one straight length, speed and velocity agree. Back at the starting wall the displacement is zero, so the average velocity is zero however hard the swimmer worked. Four significant figures come from the times; the distances are exact.

2. Hallway walk, east positive.

   (a) Leg 1: $\dfrac{+8.0\text{ m}}{4.0\text{ s}} = \mathbf{+2.0\ m/s}$ Leg 2: $\dfrac{-5.0\text{ m}}{5.0\text{ s}} = \mathbf{-1.0\ m/s}$ Leg 3: $\dfrac{-7.0\text{ m}}{5.0\text{ s}} = \mathbf{-1.4\ m/s}$

   (b) Total time $= 4.0 + 5.0 + 5.0 = 14.0$ s; distance $= 8.0 + 5.0 + 7.0 = 20.0$ m; displacement $= 8.0 - 5.0 - 7.0 = -4.0$ m.

   Average speed $= \dfrac{20.0\text{ m}}{14.0\text{ s}} = 1.428\ldots \approx \mathbf{1.4\ m/s}$ Average velocity $= \dfrac{-4.0\text{ m}}{14.0\text{ s}} = -0.2857\ldots \approx \mathbf{-0.29\ m/s\ (west)}$

   The walk went back over its own path, so most of the distance cancels in the displacement. Accept 1.43 m/s and -0.286 *m/s. Do not accept averaging the three leg velocities:* $(2.0 - 1.0 - 1.4) \div 3$ *ignores that the legs take different times.*

3. Frecciarossa.

   (a) The speed is wanted in km/h, so first write the time in hours: 2 h 59 min $= 2 + \dfrac{59}{60} = 2.98333\ldots$ h.

   $\bar{v} = \dfrac{570\text{ km}}{2.98333\ldots\text{ h}} = 191.06\ldots \approx \mathbf{191\ km/h}$

   Accept 190 km/h: "about 570 km" supports two or three figures. Do not accept $570 \div 2.59$*: 2 h 59 min is not 2.59 h.*

   (b) **Instantaneous.** *The display shows the speed at that moment. The average is lower because the train starts from rest in Rome, slows to stop in Milan, and runs more slowly on the approaches to both cities and on curves. Any one sound reason earns the mark.*

4. Instantaneous velocity.

(a)

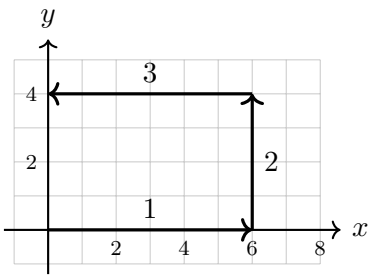
Interval	Δt (s)	Δx (m)	$\bar{v}$ (m/s)
1.00 s to 2.00 s	1.00	0.750	**0.750**
1.00 s to 1.50 s	0.50	0.313	**0.626**
1.00 s to 1.10 s	0.10	0.053	**0.53**
1.00 s to 1.05 s	0.05	0.026	**0.52**

(b) **About 0.50 m/s.** *The averages fall toward 0.5 as the interval shrinks. The data were made from $x = 0.25\,t^2$, for which the exact value at $t = 1.00$ s is 0.500 m/s.*

(c) *Positions are recorded to the nearest millimeter. Over 0.05 s the cart moves only 26 mm, so Δx has just two significant figures; over 0.01 s it would move about 5 mm, one significant figure, and $\bar{v}$ would be no better than "about 0.5". The measurement precision runs out before the interval does. Also accept: the data only has points every 0.05 s.*

5. Soccer drill, east $+x$, north $+y$.

(a) Path drawn head to tail: $(0, 0) \to (6, 0) \to (6, 4) \to (0, 4)$.



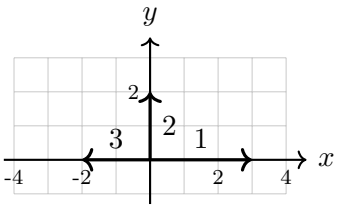
path: 1 square = 1 m

(b) Leg 1: $\dfrac{6.0 \text{ m}}{2.0 \text{ s}} = \mathbf{3.0\ m/s\ east}$

Leg 2: $\dfrac{4.0 \text{ m}}{2.0 \text{ s}} = \mathbf{2.0\ m/s\ north}$

Leg 3: $\dfrac{6.0 \text{ m}}{3.0 \text{ s}} = \mathbf{2.0\ m/s\ west}$

Legs 2 and 3 have the same speed but different velocities: the arrows point different ways.



velocity: 1 square = 1 m/s

(c) Distance $= 6.0 + 4.0 + 6.0 = 16.0$ m in 7.0 s.

Average speed $= \dfrac{16.0 \text{ m}}{7.0 \text{ s}} = 2.285\ldots \approx$ **2.3 m/s**

The player finishes at $(0, 4)$, so the displacement is 4.0 m north.

Average velocity $= \dfrac{4.0 \text{ m}}{7.0 \text{ s}} = 0.5714\ldots \approx$ **0.57 m/s north**

Reading the finish point off the grid is all that is needed here; adding vectors in two dimensions comes later. Accept 2.29 m/s and 0.571 m/s north. The average velocity is small because the player ends only 4 m from the start.